

15.1. Introduction

The word probability denotes chance and the theory of probability deals with laws governing the chances of occurrence of phenomena, which are unpredictable in nature. The theory of probability was first developed in the seventh century when certain gambling games were analysed by the French Mathematician Blaise Pascal. Probability theory today has applications for beyond games of chance, its utility is in almost all branches of Science, Engineering, Economics, Business, Industry. The Russian Mathematician A. N. Kolmogorov introduced the new approach to the concept of probability through the set theory.

In day to day life we make statements like :

1. Tomorrow will probably be a Sunny Day.
2. Most probably I shall be returning within a week.
3. It is almost impossible to drive from here to Kolkata within 4 Hours.

In all those statements there is a lack of certainty. The theory of probability plays a vital role in making decisions in situation where there is a lack of certainty. Discrete probability is concerned with one number or a few numbers that are associated with the outcome of experiments. Before discussing the theory of probability we shall explain some basic terms required in the study of probability.

15.2. Random Experiment

An operation which can produce some well defined outcomes is known as experiment. For example, tossing a coin is an experiment because if a coin is tossed either head or tail will turn up i.e., we have two possible outcomes. Similarly, drawing a ball from an urn, recording the diameter of a bolt or noting the quality of a bulb whether defective or non-defective are all experiments besides the more sophisticated investigations that are conducted in a laboratory.

Any experiment whose outcome can not be determined in advance is a random experiment.

Examples of random experiments are:

1. Tossing a coin
2. Throwing a die
3. Aiming a target.

In all these cases there are number of possible results which can occur but the actual result that will occur can not be determined in advance.

15.3. Sample Space

The set of all possible outcomes in a random experiment is called a sample space. Each element of a sample space is called a **sample point**. The number of sample points in S may be denoted by $n(S)$.

The examples of sample space and sample points are:

1. In the throwing of a die, the sample space $S = \{1, 2, 3, 4, 5, 6\}$, $n(s) = 6$ and typical sample points are 2 and 5.

2. In tossing a coin the sample space $S = \{\text{Head, Tail}\}$, $n(s) = 2$.

3. When three coins are thrown simultaneously the sample space

$$S = \{\text{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}\}, n(S) = 8.$$

4. When two die are thrown simultaneously, sample space consists of 36 sample points.

$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

where 1st number in any bracket denote the score on 1st die and 2nd number the score on 2nd die

Events

The results (or outcomes) of experiments are called events. So, an event is a subset of sample space. For example, for the sample space in tossing two coins, the subset $A = \{HH, HT, TH\}$ is the event that at least one head occurs. The subset $B = \{TT\}$ for the sample space of throwing three coins is the event of occurring three tails. The number of sample points in an event E is denoted by $n(E)$.

Simple (Elementary) event

An event consisting of single sample point is called a simple event. The six events in S of throwing a die are $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$. In a simultaneous toss of two coins, the event $\{HH\}$ of getting both heads is a simple event where $S = \{HH, TH, HT, TT\}$.

Impossible event

An Impossible event is an event which will never occur, i.e., the event that includes no elementary events at all. The event of getting three heads in tossing of two coins is an impossible event.

Sure event

An event which is sure to occur is called a sure event. In throwing a die an event consisting of number lying between 1 and 6 is a sure event.

Equally likely events

A number of events are said to be equally likely if any one of them can not be expected to occur in preference to the other. In tossing a fair coin, the two possible outcomes Head and Tail are equally likely i.e., we have no reason to accept that Heads will appear more often than Tails or vice-versa. Similarly, in a throw of an unbiased die the coming up of 1, 2, 3, 4, 5 and 6 is equally likely.

Exhaustive events

Events are said to be exhaustive when they include all possible outcomes of a random experiment. In tossing a coin, exhaustive events are two. In rolling a die, there are six exhaustive cases, since any one of the six numbers may appear on top.

15.4. Set Notations

Since the events are sets, the statements concerning events can be written using set notations. If A and B are two events then

- (i) The union $A \cup B$ is the event of happening at least one of the events A and B .
- (ii) The difference $A - B$ means the occurrence of the event A with non - occurrence of the event B .
- (iii) The intersection $A \cap B$ is the event of both events A, B happening.
- (iv) A' is the event which occurs if and only if A does not occur.

Example 1. Let A be the event that a business executive, selected at random has stomach ulcer and B be that he has blood pressure, then

$A \cup B$ is the event that he has at least one of diseases, stomach ulcer or blood pressure.

$A - B$ is the event that he has stomach ulcer.

$A \cap B$ is the event he has both and

A' is the event that he does not have stomach ulcer.

Example 2. If A be the event of getting an even number in the throw of a die and B be the event of getting a number less than 4, then $A = \{2, 4, 6\}$ and $B = \{1, 2, 3\}$ then
 $A \cup B = \{1, 2, 3, 4, 6\}$ is the event of getting an even number or a number less than 4.
 $A - B = \{4, 6\}$ is the event of getting even number not included in B.
 $A \cap B = \{2, 3\}$ is the event of getting even number less than 4 and
 $A' = \{1, 4, 5\}$ is the event of not getting even number.

Mutually Exclusive Events

Two or more events are mutually exclusive or disjoint events if the events cannot occur simultaneously, in other words, occurrence of one of the events prevents the occurrence of others. If A and B are any two events defined on a sample space S and $A \cap B = \Phi$, i.e., there is no common sample points between them, then the events A and B are said to be mutually exclusive. For example, in tossing a coin the appearance of head and tail are mutually exclusive events as they can not occur simultaneously.

Probability of Events

Every event associated with a random experiment is assigned a weight or measure of the chance of it occurring called its **probability**. The probability of an occurrence of an event A in a sample S written as $P(A)$, is defined by

$$P(A) = \frac{\text{Number of distinct sample points (elements) in } A}{\text{Number of distinct sample points (elements) in } S} = \frac{n(A)}{n(S)}$$

The number of distinct sample points in A are called the number of outcomes favourable to the occurrence of A and number of distinct sample points in S are called the total number of possible outcomes. Then,

$$P(A) = \frac{\text{Number of outcomes favourable to the occurrence of } A}{\text{Total number of all possible outcomes}}$$

The probability of non-occurrence of the event A is

$$\begin{aligned} P(A') &= \frac{\text{Number of unfavourable outcomes}}{\text{Total number of all possible outcomes}} \\ &= \frac{n(S) - n(A)}{n(S)} = 1 - \frac{n(A)}{n(S)} = 1 - P(A) \end{aligned}$$

or $P(A) + P(A') = 1$

Axioms of Probability

The probability of an event A of sample space S must satisfy the following axioms:

1. Positiveness: $0 \leq P(A) \leq 1$ for every event A of S.

2. Certainty: $P(S) = 1$

3. Union: $P(A \cup B) = P(A) + P(B)$ for any two events A and B of S for which $A \cap B = \Phi$

If $A_1, A_2, \dots$ is a sequence of mutually exclusive events, then

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

All of conventional probability theory follows from three axioms of probability.

Addition Rule

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For three events A, B and C, applying the above rule twice, we obtain

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

The addition rule can be generalized as given below

If $A_1, A_2, \dots, A_n$ are any events, then:

$$P\left(\bigcup_{i=1}^n A_i\right) = P(A_1 \cup A_2 \cup \dots \cup A_n) \\ = \sum_i P(A_i) - \sum_{i < j \leq n} P(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) + \dots \\ + (-1)^{n-1} P(A_1 \cap A_2 \cap \dots \cap A_n),$$

where the successive sums are over all possible events, pairs of events, triples of events, and so on.

Solved Examples

Type I (on finding sample space and event).

Example 3. There are 4 red and 3 blue balls in a bag. If one ball is taken out of this bag at random, then represent the sample space and the event of this ball being blue and also find the number of elements in them.

Solution. Let R and B denote Red and Blue balls respectively.

Thus, the required sample space $S = \{R, R, R, R, B, B, B\}$

Then $n(S) =$ The number of sample points in $S = 7$

Let E be the event of getting a blue ball, $E = \{B, B, B\}$

Then $n(E) =$ Then number of sample points in $E = 3$.

Example 4. If three coins are tossed, represent the sample space and the event of getting (i) two heads and one tail (ii) at least two tails and the number of sample points in them.

Solution. Let S be the sample space.

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

where HHT denotes head on the first coin, head on the second coin and tail on the third. So, $n(S) = 8$.

(i) Let E_1 be the event of getting two heads and one tail.

So, $E_1 = \{HHT, HTH, THH\}$ and $n(E_1) = 3$.

(ii) Let, E_2 be the event of getting two tails and more.

Then, $E_2 = \{HTT, THT, TTH, TTT\}$ and $n(E_2) = 4$.

Example 5. State which of the following events are mutually exclusive. Give reasons for your answer.

(i) E_1 : a black card, E_2 : an ace, in a draw of a card from a deck.

(ii) E_1 : a number less than five, E_2 : an even number in a single throw of 2 dice.

(iii) E_1 : a boy, E_2 : a girl, when a child is born.

(iv) E_1 : a spade, E_2 : a diamond, in a draw of a card from a deck.

Solution. (i) E_1 and E_2 are not mutually exclusive events as black ace is favourable to both events.

(ii) E_1 and E_2 are not mutually exclusive events as (1, 2), (3, 2) are events which are favourable to both E_1 and E_2 .

(iii) E_1 and E_2 are mutually exclusive events as a child can not be both a boy and a girl.

(iv) E_1 and E_2 are mutually exclusive events as a card can not be both a spade and a diamond.

Type II (on direct application of $P(E) = \frac{n(E)}{n(S)}$)

Example 6. Two unbiased dice are thrown. Find the probability that

- getting a sum of 6
- the numbers shown are equal
- the difference of the numbers shown is 1
- the first die shows 6
- the total of numbers greater than 8.

Solution. The two dice can be thrown in 6×6 ways = 36 ways

So, $n(S) = 36$ where S is the sample space

(i) Let, E_1 be the events of getting a sum of 6.

So, $E_1 = \{(1, 5), (5, 1), (2, 4), (4, 2), (3, 3)\}$ and $n(E_1) = 5$.

Hence, the required probability = $\frac{n(E_1)}{n(S)} = \frac{5}{36}$

(ii) Let, E_2 be the events of getting equal numbers.

So, $E_2 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$ and $n(E_2) = 6$.

Hence, the required probability = $\frac{n(E_2)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

(iii) Let, E_3 be the events of showing the difference of numbers is 1.

So, $E_3 = \{(1, 2), (2, 1), (3, 2), (2, 3), (3, 4), (4, 3), (4, 5), (5, 4), (5, 6), (6, 5)\}$ and $n(E_3) = 10$.

Hence, the required probability = $\frac{n(E_3)}{n(S)} = \frac{10}{36} = \frac{5}{18}$

(iv) Let, E_4 be the event of first die showing 6.

So, $E_4 = \{(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$ and $n(E_4) = 6$

Hence, the required probability = $\frac{n(E_4)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

(v) Let, E_5 be the event of the total numbers greater than 8.

So, $E_5 = \{(3, 6), (6, 3), (4, 5), (5, 4), (5, 5), (4, 6), (6, 4), (5, 6), (6, 5), (6, 6)\}$ and $n(E_5) = 10$.

Hence, the required probability = $\frac{n(E_5)}{n(S)} = \frac{10}{36} = \frac{5}{6}$

Example 7. Three unbiased coins are tossed.

(a) Write the sample space S

(b) Find the probability of

- all heads
- at least 2 heads
- at most 2 heads.

Solution. (a) Let S denotes the sample space of tossing three coins.

$\therefore S = \{HHH, HHT, HTH, THH, HTT, TTH, THT, TTT\}$ and hence $n(S) = 8$

(b) (i) Let, E_1 be the set of favourable events then

$E_1 = \{HHH\}$ and $n(E_1) = 1$

$\therefore P(E_1) = \frac{n(E_1)}{n(S)} = \frac{1}{8}$

(ii) Let E_2 be the set of favourable events then

$E_2 = \{HHT, HTH, THH, HHH\}$ and $n(E_2) = 4$

$\therefore P(E_2) = \frac{4}{8} = \frac{1}{2}$

(iii) Let, E_3 be the set of favourable events

$\therefore E_3 = \{\text{HHT, HTH, THH, HTT, TTH, THT, TTT}\}$ and $n(E_3) = 7$

$$\therefore P(E_3) = P(\text{at most 2 heads}) = \frac{n(E_3)}{n(S)} = \frac{7}{8}$$

Type III (Using permutation and combination)

Example 8. A bag contains 7 red and 4 white balls. Two balls are drawn at random. Find the probability that

(a) both the balls are red (b) one ball is white and another one is red.

Solution. The number of ways of selecting 2 balls out of total 11 balls is

$$C(11, 2) = \frac{11 \times 10}{2} = 55$$

$\therefore n(S) = 55$ where S is the sample space.

(a) Let, E_1 be the event that the selected balls are red. The number of ways of selecting 2 red balls out of 7 red balls is

$$C(7, 2) = \frac{7 \times 6}{2} = 21$$

$$\therefore n(E_1) = 21$$

$$\therefore \text{The required probability} = P(E_1) = \frac{n(E_1)}{n(S)} = \frac{21}{55}$$

(b) Let, E_2 be the event that one ball is white and another one is red. The number of ways of selecting one white ball out of 4 white balls and one red ball out of 7 red balls is $C(4, 1) \times C(7, 1) = 4 \times 7 = 28$.

$$\therefore \text{The required probability} = P(E_2) = \frac{n(E_2)}{n(S)} = \frac{28}{55}$$

Example 9. A committee of 5 Principals is to be selected from a group of 6 gent principals and 8 lady Principals. If the selection is made randomly, find the probability that there are 3 lady principals and 2 gent Principals.

Solution. The number of ways that 5 Principals can be selected from total 14 Principals is $C(14, 5)$

$$= \frac{14 \times 13 \times 12 \times 11 \times 10}{5 \times 4 \times 3 \times 2} = 14 \times 13 \times 11.$$

The number of ways of selecting 3 lady Principals from 8 lady Principals and 2 gent Principals from 6 gent Principals

$$= C(8, 2) \times C(6, 2) = \frac{8 \times 7 \times 6}{3!} \times \frac{6 \times 5}{2!} = 8 \times 7 \times 15$$

$$\therefore \text{The required probability} = \frac{8 \times 7 \times 15}{14 \times 13 \times 11} = \frac{60}{143}$$

Example 10. Three electric bulbs are chosen at random from 15 bulbs of which 5 are defective. Find the probability that (i) none is defective, (iii) exactly one is defective (iii) at least one is defective.

Solution. Three bulbs can be chosen from 15 bulbs in

$$C(15, 3) = \frac{15 \times 14 \times 13}{3 \times 2} = 455 \text{ ways}$$

Again, the number of ways of selecting 3 non-defective bulbs from $(15 - 5)$ i.e., 10 non-defective bulbs = $C(10, 3) = 180$.

$$\therefore \text{The probability that none is defective} = \frac{120}{455} = \frac{24}{91}$$

(ii) the number of ways of selecting one defective from 5 defective bulbs and 2 non-defective from 10 non-defective = $C(5, 1) \times C(10, 2) = 5 \times 45 = 225$.

$$\therefore \text{The required probability} = \frac{225}{455} = \frac{45}{91}$$

$$\begin{aligned} \text{(iii) Probability of at least one defective} &= 1 - \text{Probability of no defective} \\ &= 1 - \frac{24}{91} = \frac{67}{91} \end{aligned}$$

Type V (On mutually exclusive events)

Example 11. The probability of a horse A winning a race is $1/5$ and the probability of a horse B winning the same is $1/4$. What is the probability that (i) either of them will win (ii) none of them will win?

Solution. Let C and D be the events of winning the race by horse A and B respectively.

$$\text{Given } P(C) = 1/5 \text{ and } P(D) = 1/4$$

(i) The event of either A or B winning is $C \cup D$ where C and D are mutually exclusive since both of them can not win together. So, the probability of any one of them winning

$$P(C \cup D) = P(C) + P(D) = 1/5 + 1/4 = 9/20$$

(ii) Clearly, the event of none of the horses winning is $(C \cup D)'$

$$\text{So, } P(C \cup D)' = 1 - P(C \cup D) = 1 - 9/20 = 11/20$$

Example 12. The lifespans of 5000 electrical components are measured to assess their reliability. The lifespan (L) is recorded and the results are shown in Table 15.1 Find the probability that a randomly selected component will last

- (i) more than 4 years
- (ii) between 4 and 6 years
- (iii) less than 5 years

Lifespan of component (yrs)	Number
$L > (6)$	600
$5 < L \leq 6$	2250
$4 < L \leq 5$	1850
$L \leq 4$	300

Table 15.1. The lifespan of 5000 electrical components.

Solution. We define the events A, B, C and D as follows

- A : the component lasts more than 6 years
- B : the component lasts between 5 and 6 years
- C : the component lasts between 4 and 5 years
- D : the component lasts 4 years or less

$$P(A) = 600/5000 = 0.12, P(B) = 2200/5000 = 0.44, P(C) = 1900/5000 = 0.38 \text{ and } P(D) = 300/5000 = 0.06. \text{ The events A, B, C and D are clearly mutually exclusive. Hence}$$

$$(i) P(\text{component lasts more than 4 years}) = P(A \cup B \cup C) = P(A) + P(B) + P(C) = 0.12 + 0.44 + 0.38 = 0.94$$

There is a 94 % chance a component will last for more than 4 years.

$$(ii) P(\text{component lasts between 4 and 6 years}) = P(B \cup C) = P(B) + P(C) = 0.44 + 0.38 = 0.82$$

$$(iii) P(\text{components lasts less than 5 years}) = P(C \cup D) = P(C) + P(D) = 0.38 + 0.06 = 0.44$$

Type VI (on Addition Rule)

Example 13. A drawer contains 50 bolts and 150 nuts. Half of the bolts and half of the nuts are rusted. If one item is chosen at random, what is the probability that it is rusted or a bolt.

Solution. Let A be the event of drawing a rusted item and B be the event of drawing a bolt. Now, total number of bolts is 50 and number of rusted bolts is 25. Total number of nuts is 150 and number of rusted nuts is 75. So, total of items is $150 + 50 = 200$ and total number of rusted items is $25 + 75 = 100$. The events A and B are not mutually exclusive events since some bolts are rusted.

$$\begin{aligned} P(A \cap B) &= P(\text{drawing rusted nuts}) \\ &= 25/200 = 1/8 \end{aligned}$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 1/2 + 1/4 - 1/8 = 5/8. \end{aligned}$$

Example 14. The probability that a contractor will get a plumbing contract is $2/3$ and the probability that he will not get an electric contract is $5/9$. If the probability of getting at least one contract is $4/5$, What is the probability that he will get both the contracts ?

Solution. Let A be the event that the contractor will get the plumbing contract and B, that he will get the electric contract. Then

$$P(A) = \frac{2}{3}, P(B') = \frac{5}{9} \text{ and } P(A \cup B) = \frac{4}{5}.$$

We have to find the value of $P(A \cap B)$.

$$\text{We have } P(B) = 1 - P(B') = 1 - \frac{5}{9} = \frac{4}{9}.$$

By addition rule, we have $P(A \cup B) = P(A) + P(B) - P(A \cap B)$,

$$\therefore \frac{4}{5} = \frac{2}{3} + \frac{4}{9} - P(A \cap B) = \frac{10}{9} - P(A \cap B),$$

$$\text{or, } P(A \cap B) = \frac{10}{9} - \frac{4}{5} = \frac{50 - 36}{45} = \frac{14}{45}.$$

$$\text{Hence, the required probability} = \frac{14}{45}.$$

15.6. Compound Event

When two or more events occur in conjunction with each other, their joint occurrence is called compound event. For example,

(i) If a die is rolled two times, the event of coming up of 1 in the first throw and 6 in the second throw is a compound event.

(ii) If four cards are drawn in succession from a pack of cards, the drawing of the cards from the same suit all the four times is a compound event.

The compound events are further classified into (i) Independent events (ii) Dependent events.

Independent Events

Events are said to be independent if the occurrence of one of them does not affect the occurrence of any of the other. For examples,

- (i) In tossing two coins, the appearance of head on one coin does not affect the appearance of head on the other coin.
- (ii) If one ball is taken out from a bag containing 6 balls and is kept again in the bag. Next, another ball is drawn, then drawing of the first and second balls are independent events.

Dependent Events

If the happening of one event influences the probability of the happening of the other (these events being mutually exclusive) then the second event is said to be dependent on the first. For examples,

- (i) If A draws a card from a full pack and does not replace it, the draw afterward made by B is dependent on the draw made by A.
- (ii) If one ball is taken out from a bag containing 6 balls and a second ball is drawn without replacing the first. Then the draw of the second depends upon what was drawn first. So, the events are dependent events.

15.7. Conditional Probability

Let A be an arbitrary event in a sample space S with $P(A) > 0$. The probability that an event B occurs after A has occurred is called the conditional probability of B given A and is written as $P(B/A)$ and is defined as

$$P(B/A) = \frac{P(A \cap B)}{P(A)} \quad (1)$$

Similarly, $P(A/B)$ defines the conditional probability of A when B has already occurred.

Product Rule

If in formula (1), we multiply both sides of the equation by $P(A)$, we obtain the following useful relationship, which is referred to as the product rule:

$$P(A \cap B) = P(A) P(B/A)$$

If A and B are independent events, then

$$P(A/B) = P(A) \quad \text{and} \quad P(B/A) = P(B)$$

Hence $P(A \cap B) = P(A) P(B)$

Note (i) Three events A, B, C are said to be mutually independent if

$$P(A \cap B) = P(A) P(B)$$

$$P(A \cap C) = P(A) P(C)$$

$$P(B \cap C) = P(B) P(C), \text{ and}$$

$$P(A \cap B \cap C) = P(A) P(B) P(C)$$

In case the first three conditions are satisfied and the fourth is not satisfied then A, B, C are said to be pairwise independent.

Example 15. What is the conditional probability that a randomly generated bit string of length four contains at least two consecutive 0s, given that the first bit is a 1? (Assume the probabilities of a 0 and a 1 are the same)

Solution. Let E be the event that a bit string of length four contains at least two consecutive 0s and F be the event that the first bit is a 1. The probability that a bit string of length four has at least two consecutive 0s, given that its first bit is a 1 equals $P(E/F)$.

Without restriction the number of ways of bit string of length four can be formed with digits 0 and 1 is $2^4 = 16$. Keeping 1 fixed at the beginning, there are 8 (2^3) bit strings of length four that start with a 1. Thus $P(F) = 8/16 = 1/2$. Again $E \cap F = \{1000, 1100, 1001\}$, so $P(E \cap F) = 3/16$. Consequently

$$P(E/F) = \frac{P(E \cap F)}{P(F)} = \frac{3/16}{8/16} = 3/8.$$

Example 16. The personal department of a company has records which show the following analysis of its 200 engineers:

Age	Bachelors degree only	Masters degree	Total
Under 30	90	10	100
30 to 40	20	30	50
Over 40	40	10	50
Total	150	50	200

If one engineer is selected at random from the company, find:

- The probability that he has only a bachelor's degree.
- The probability that he has a master's degree, given that he is over 40.
- The probability that he is under 30, given that he has only a bachelor's degree.

Solution. We define the events A, B, C and D as follows:

A: an engineer has a bachelor's degree

B: he is over 40 years of age.

C: he has master's degree

D: he is under 30 years of age.

- The probability that he has only a bachelor's degree is $P(A) = 150/200 = 3/4$.
- The probability that an engineer has a master's degree, given that he is over 40 is

$$P(C/B) = \frac{P(C \cap B)}{P(B)} = \frac{10/200}{50/200} = 10/50 = 1/5$$

- The probability that an engineer is under 30, given that he has only a bachelor's degree is

$$P(D/A) = \frac{P(D \cap A)}{P(A)} = \frac{90/200}{150/200} = 90/150 = 3/5$$

Example 17. A problem in Discrete Mathematics is given to three students whose chances of solving it are $1/2$, $1/3$, $1/4$ respectively. What is the probability that only one of them solves it correctly?

Solution. Let A be the event that student A will solve the problem. Similarly B and C be the events for the students B and C respectively, to solve the problem.

Then $P(A) = 1/2$, $P(B) = 1/3$, $P(C) = 1/4$.

$P(A^c)$ = Probability that A will not solve the problem.

$$= 1 - P(A) = 1 - 1/2 = 1/2.$$

Similarly, $P(B^c) = 1 - 1/3 = 2/3$, and $P(C^c) = 1 - 1/4 = 3/4$.

The probability that none of the students A, B, C, will solve the problem $= P(A^c) \cdot P(B^c) \cdot P(C^c) = 1/2 \times 2/3 \times 3/4 = 1/4$.

[A, B, C are independent events, A^c , B^c , C^c are also independent events.]

Hence, the required probability that the problem will be solved (i.e., at least one of A, B, C will solve the problem) $= 1 - 1/4 = 3/4$.

Theorem of Total Probability for compound Events

If $A_1, A_2, \dots, A_n$ be subsets of a sample space S such that

- (i) Each A_i is a proper subset of S i.e., $A_i \subset S$, $i = 1, 2, \dots, n$ and $A_i \neq S$
 (ii) $A_1 \cup A_2 \cup \dots \cup A_n = S$
 (iii) The subsets A_i are pairwise disjoint, that is
 $A_i \cap A_j = \Phi$, $i, j = 1, 2, 3, \dots, n$ $i \neq j$

If B is any event, then

$$P(B) = \sum_{i=1}^n P(A_i) P(B/A_i) \text{ provided } P(A_i) \neq 0, i = 1, 2, \dots, n$$

Proof. By (ii), we have $S = \bigcup_{i=1}^n A_i$

$$\text{Now } B = B \cap S = B \cap \left(\bigcup_{i=1}^n A_i \right) = \bigcup_{i=1}^n (B \cap A_i)$$

[by distributive law]

Since $A_i \cap A_j = \Phi$ for $i \neq j$, it follows that
 $(B \cap A_i) \cap (B \cap A_j) = \Phi$ for $i \neq j$

$$\text{Hence } P(B) = P \left[\bigcup_{i=1}^n (B \cap A_i) \right] = \sum_{i=1}^n P(B \cap A_i)$$

[by additive theorem]

$$= \sum_{i=1}^n P(A_i) P(B/A_i)$$

[by product theorem]

Baye's Theorem

If $A_1, A_2, \dots, A_n$ are n mutually exclusive and exhaustive events in a sample space S and B is another event in S then

$$\begin{aligned} P(A_k/B) &= \frac{P(A_k) P(B/A_k)}{P(A_1) P(B/A_1) + P(A_2) P(B/A_2) + \dots + P(A_n) P(B/A_n)} \\ &= \frac{P(A_k) P(B/A_k)}{\sum_{i=1}^n P(A_i) P(B/A_i)} \end{aligned}$$

Proof. By theorem of total probability, we have

$$P(B) = \sum_{i=1}^n P(A_i) P(B/A_i) \quad \dots (1)$$

$$\text{Also } P(B \cap A_k) = P(A_k) P(B/A_k) \quad \dots (2)$$

$$\begin{aligned} \text{Hence } P(A_k/B) &= \frac{P(B \cap A_k)}{P(B)} \\ &= \frac{P(A_k) P(B/A_k)}{\sum_{i=1}^n P(A_i) P(B/A_i)} \text{ by (1) and (2)} \end{aligned}$$

The probabilities $P(A_1), P(A_2), \dots, P(A_n)$ which are already known before conducting experiment are termed as **Priori probabilities** and the conditional probabilities $P(A_i/B)$,

$P(A_2/B), \dots, P(A_n/B)$ which are computed after conducting the experiment are termed as **Posteriori probabilities**.

Example 18. Suppose there are three urns containing 2 white and 3 black balls ; 3 white and 2 black, and 4 white and 1 black balls respectively. There is equal probability of each urn being chosen. One ball is drawn from an urn chosen at random. What is the probability that a white ball is drawn ?

Solution. Let A_i be the event that i th urn is chosen, $i = 1, 2, 3$ and $P(A_i) = 1/3$ and B the event that a white ball is drawn.

Then
$$P(B) = P(A_1) P(B/A_1) + P(A_2) P(B/A_2) + P(A_3) P(B/A_3)$$

Here
$$P(A_1) = P(A_2) = P(A_3) = 1/3$$

and
$$P(B/A_1) = 2/5, P(B/A_2) = 3/5, P(B/A_3) = 4/5.$$

$$P(B) = 1/3 (2/5 + 3/5 + 4/5) = 3/5$$

Example 19. Three machines A, B and C produce respectively 60%, 30% and 10% of the total number of items of a factory. The percentages of defective output of these machines are respectively 2%, 3% and 4%. An item is selected at random from the product and is found defective. Find the probability that the item was produced by machine C.

Solution: Let, A_1, B_1, C_1 be the events that item drawn at random is produced by machines A, B, C respectively and let X be the event that the product drawn is defective.

Then $P(A_1)$ = Probability that the product drawn was manufactured by A

$$= \frac{60}{100} = .60$$

Similarly $P(B_1) = \frac{30}{100} = .30$ and $P(C_1) = \frac{10}{100} = .10.$

$P(X/A_1)$ = Probability of drawing a defective item from the product manufactured by A

$$= \frac{2}{100} = .02$$

Similarly $P(X/B_1) = \frac{3}{100} = .03$ and $P(X/C_1) = \frac{4}{100} = .04$

We seek $P(C_1/X)$ = Probability that an item is produced by Machine C given that the item is defective.

By Baye's theorem:
$$P(C_1/X) = \frac{P(C_1) P(X/C_1)}{P(A_1) P(X/A_1) + P(B_1) P(X/B_1) + P(C_1) P(X/C_1)}$$
$$= \frac{(0.10) (0.04)}{(0.60) (0.02) + (0.30) (0.03) + (0.10) (0.04)}$$
$$= \frac{4}{25}.$$

Example 20. In a test, an examinee either guesses or copies or knows the answer to multiple-choice question with four choices, only one answer being correct. The probability that he makes a guess is $\frac{1}{3}$ and the probability that he copies the answer is $\frac{1}{6}$. The probability that his answer is correct, given that he copies it, is $\frac{1}{8}$. Find the probability that he knew the answer to the question, given that he correctly answers it.

Solution. Let A: the event of answering by guess work
 B: the event of answering by coping
 C: the event of answering by knowing
 D: the event of answering correctly.

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{6}$$

Then
 As the question is answered by either guessing or copying or knowing,

$$P(C) = 1 - P(A) - P(B)$$

$$P(C) = 1 - \frac{1}{3} - \frac{1}{6} = \frac{1}{2}$$

Now, $P(D/A)$ = Probability of answering correctly by guessing.

$$= \frac{1}{4} \{ \because \text{there are 4 choices and only one of them is correct} \}$$

$P(D/B)$ = Probability of answering correctly by copying.

$$= \frac{1}{8} \text{ (given)}$$

$P(D/C)$ = Probability of answering correctly by knowing.

$$= 1 \quad (\because \text{If he knows, he certainly answer correctly}).$$

We have to find the probability that he knew when he answered correctly i.e., $P(C/D)$

By Bayes' theorem, $P(C/D) = \frac{P(C).P(D/C)}{P(A).P(D/A) + P(B).P(D/B) + P(C).P(D/C)}$

$$= \frac{\frac{1}{2} \cdot 1}{\frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot \frac{1}{8} + \frac{1}{2} \cdot 1}$$

$$= \frac{\frac{1}{2}}{\frac{1}{12} + \frac{1}{48} + \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{29}{48}} = \frac{24}{29}$$

15.8. Random Variable

If an experiment of tossing a coin is performed twice and X represents the number of heads which can come up, then we can associate a number for X as shown in the table below.

Sample point	HH	HT	TH	TT
X	2	1	1	0

Thus X takes the values 2, 1, 0 corresponding to two heads, one head or no head and X is called a random variable.

A variable whose numerical value is determined by the outcome of a random experiment is called a random variable. A random variable X is a real valued function, $X(x)$, of the elements of

the sample space S where x is an element of the sample space. It should be noted that the range of the random variable will be a set of real numbers. For examples,

(i) If we flip a coin and denote the head by 1 and tail by 0, then the random variable X takes only two values 1 and 0. Symbolically, the random variable,

$$X(x) = \{x : x = (1, 0) \in S\}$$

(ii) If we measure the height of people, the random variable,

$$X(x) = \{x : x \text{ is any real positive number}\}$$

Random variables are of two types (i) Discrete random variable and (ii) Continuous random variable.

Discrete Random Variable: a random variable X , which can take only a finite number of values in an interval of the domain, is called discrete random variable. A few examples are:

(i) If we throw two dice at a time and note the sum of spots which turn up on the upper face, the discrete random variable,

$$X(x) = \{x : x = (x_1 + x_2) = 2, 3, \dots, 12\}$$

where x_1 is the number of spots on the upper face of one die and x_2 is the number of spots on the other die

(ii) The random variable denoting the number of students in a class is,

$$X(x) = \{x : x \text{ is any positive integer}\}$$

Continuous Random Variable: A random variable X , which can take any value in the domain, or when its range 'R' is an interval or the union of intervals on the real line, is called a continuous random variable.

Note that the probability of any single x , a value of X , is zero, i.e., $P(X = x) = 0$.

Example (i) The height of students in India lies between 3cm and 6cm. The continuous random variable,

$$X(x) = \{x : 3 \leq x \leq 6\}$$

(ii) The maximum life of electric bulbs is 3000 hours. The continuous random variable,

$$X(x) = \{x : 0 \leq x \leq 3000\}.$$

15.9. Discrete Probability Distribution

Probability distribution of a random variable is the set of its possible values together with their respective probabilities. For a discrete random variable (X , say) the possible values can be written as a list $\{x_1, x_2, x_3, \dots\}$ corresponding probabilities $P(X = x_1)$, $P(X = x_2)$, $P(X = x_3)$, The occurrence of the values x_i with their probabilities is called the discrete probability distribution of X .

Discrete Probability Function Often, the probability (p) that the random variable X assumes a specified value (x) that can be expressed in terms of general mathematical expression $f(x)$

$$\begin{aligned} f(x) &= \text{Probability that } X \text{ assumes the value } x \\ &= P(X = x) \end{aligned}$$

The function $f(x)$ is known as **probability mass function (pmf)** or **probability function** of the discrete random variable X .

Expectation – Mean, Variance

Suppose a discrete random variable X assume the values $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$ respectively. Then the expectation or expected value of X , $E(X)$, is defined as the sum of products of different values of x and the corresponding probabilities.

Symbolically,

$$E(X) = \sum p_i x_i$$

Similarly

$$E(X^2) = \sum p_i x_i^2$$

In general, the expected value of any function $g(X)$ of a random variable X is defined as

$$E[g(X)] = \sum p_i g(x_i)$$

Some results on Expectation. If X and Y are two variates defined over the sample space and $E(X)$ and $E(Y)$ exist, then

$$E(X + Y) = E(X) + E(Y)$$

(this applies whether or not X and Y are independent variable)

$$E(XY) = E(X)E(Y)$$

(this applies only when X and Y are independent)

$$E(C) = C$$

$$E(CX) = CE(X)$$

$$E(C + X) = E(X) + C \text{ where } C \text{ is a constant.}$$

If X and Y are two variables and C_1 and C_2 are any constants, then

$$E(C_1 X + C_2 Y) = C_1 E(X) + C_2 E(Y)$$

If $f(x)$ is given, the expectations are defined as follows:

$$E(X) = \sum x f(x)$$

$$E(X^2) = \sum x^2 f(x)$$

Mean of a probability distribution is the expected value of x .

$$\text{Mean } (\mu) = E(X)$$

Variance is the expected value of $(x - \mu)^2$, where μ is the mean.

$$\text{Variance } (\sigma^2) = E(x - \mu)^2$$

It may be shown that

$$\sigma^2 = E(x^2) - \mu^2$$

The positive square root of variance gives the standard deviation (σ)

Example 21. Find the probability distribution of the number of heads in two tosses of a coin.

Solution. Let X be a random variable which denoted the number of heads in two tosses of a

coin. Then

$P(X = 0) = P(TT) = 1/4$, $P(X = 1) = P(\{HT, TH\}) = 1/2$, $P(X = 2) = P(HH) = 1/4$. Hence the required probability distribution is

X	0	1	2
$P(X)$	1/4	1/2	1/4

Example 22. A random variable has the following probability distribution:

X	4	5	6	8
Probability	0.1	0.3	0.4	0.2

Find the expectation and the standard deviation of the random variable.

Solution.

$$\text{Mean } (\mu) = E(X) = \sum p_i x_i$$

$$= (0.1 \times 4) + (0.3 \times 5) + (0.4 \times 6) + (0.2 \times 8)$$

$$= 0.4 + 1.5 + 2.4 + 1.6 = 5.9$$

$$E(x^2) = \sum p_i x_i^2 = (0.1 \times 4^2) + (0.3 \times 5^2) + (0.4 \times 6^2) + (0.2 \times 8^2)$$

$$= 1.6 + 7.5 + 14.4 + 12.8 = 36.3$$

$$\text{Variance } (\sigma^2) = E(x^2) - \mu^2 = 36.3 - (5.9)^2 = 1.49$$

$$\text{Standard Deviation} = \sqrt{1.49} = 1.22$$

Example 23. The end semester exam of a discrete mathematics paper consists of 50 true/false questions, each worth two points, and 25 multiple-choice questions, each worth four points. The probability that Sheela answers a true/false correctly is 0.8, and the probability that she answers multiple choice questions correctly is 0.7. What is her expected score in the exam. ?

Solution. Let the random variable X denotes the score on True/false question and Y for the score on multiple-choice question. Hence

$$\begin{aligned} E(X) &= 0.8 \times 2 + 0.8 \times 2 + \dots \text{to 50 terms} \\ &= 80 \end{aligned}$$

$$\begin{aligned} E(Y) &= 0.7 \times 4 + 0.7 \times 4 + \dots \text{to 25 terms} \\ &= 70 \end{aligned}$$

The expected score of the sum

$$E(X + Y) = E(X) + E(Y) = 80 + 70 = 150$$

which is the expected score in the exam.

Example 24. From an urn, containing 3 red balls and 2 white, a man is to draw two balls at random without replacement, being promised Rs 20 for each red ball he draws and Rs 10 for each white one. Find his expectation.

Solution. For the possible result of the drawing there are three exhaustive and mutually exclusive cases, namely, 2 red balls, 1 red and 1 white, and 2 white. The corresponding probabilities are $3/10$, $3/5$ and $1/10$. Hence the expected value of the amount to be given is

$$\text{Rs } (3/10 \times 40 + 3/5 \times 30 + 1/10 \times 20) = \text{Rs } 32.$$

Binomial Distribution

It is one of the most popular discrete distributions. The origin of binomial distribution lies in Bernoulli's trials. A Bernoulli's trial is an experiment having only two possible outcomes, that is, success or failure. A flip of a coin (head or tail), performance of a student in a course (pass or fail), testing an item (defective or not) are examples of such experiments.

Since a Bernoulli's trial has been considered, certain conditions for the application of binomial distribution are obvious. They are

1. Same experiment is repeated several times ;
2. There are only two possible outcomes – success and failure ;
3. The repeated trials are independent ; and
4. The probability of each outcome remains the same for each trial.

The **Binomial distribution** governs the number (X , say) of success in n independent trials, with a probability p of success at each trial:

$$P(X = r) = C(n, r) p^r (q)^{n-r}$$

where $q = 1 - p$ and the range of possible values (r) is $\{0, 1, 2, \dots, n\}$.

The Binomial distribution can be thought of as the sum of n independent Bernoulli random variables. When n Bernoulli trials are carried out, the outcome is an n -tuple $(r_1, r_2, \dots, r_n)$ where $r_i = S$ (for success) or $r_i = F$ (for failure) for $i = 1, 2, \dots, n$. Since n trials are independent, the probability of each outcome of n trials consisting of r successes and $n - r$ failures is $p^r q^{n-r}$. Since there are $C(n, r)$ n -tuples of S s and F s that contain r S s, the probability of r success is $C(n, r) p^r q^{n-r}$.

This distribution has two parameters n and p . The mean of the Binomial distribution is np and variance is npq .

Example 25. If 10% bolts produced by a machine are defective, determine the probability that out of 10 bolts chosen at random (a) one will be defective, (b) at least one will be defective, and (c) at most two will be defective.

Solution. Let p and q be the probability of being defective and non defective respectively.

$$P = 10/100 = .1 \quad \text{and} \quad q = 1 - .1 = .9$$

$$(a) P(\text{exactly one defective in 10 bolts}) = P(X = 1)$$

$$= C(10, 1) (.1)^1 (.9)^9 = (.9)^{10}$$

$$(b) P(\text{at most two defectives}) = P(\text{zero defective}) + P(\text{exactly one defective})$$

$$+ P(\text{exactly two defectives})$$

$$= P(X = 0) + P(X = 1) + P(X = 2)$$

$$= C(10, 0) (.1)^0 (.9)^{10} + C(10, 1) (.1) (.9)^9$$

$$+ C(10, 2) (.1)^2 (.9)^8$$

$$= (.9)^{10} + 10 \cdot (.1) (.9)^9 + 45 (.01) (.9)^8$$

Example 26. In electronically transmitted data, errors occur randomly and independently. For transmission of binary string of length 7, find the probability of correct reception of message, if no coding were used ($p = .97$)

Using Hamming code, the receiver will get the correct message even if one is occurred. Find the probability of correct reception.

$$\text{Solution. Probability of no error} = P(X = 0) = p^7 = (.97)^7$$

When Hamming code is used, probability of correct reception

$$= P(\text{no error}) + P(\text{one error})$$

$$= P(X = 0) + P(X = 1) = p^7 + C(7, 1) p^6 q$$

$$= (.97)^7 + 7 (.97)^6 (.03)$$

Poisson Distribution

There is a need for assigning a probability to the number of occurrence of an event during a given period of time. The number of occurrences of an event, E , in a given period is a discrete random variable. Poisson distribution models the number of occurrence of an event in a given interval.

Examples of Poisson distribution are:

1. Number of printing mistakes per page in a large text book.
2. Number of particles emitted by a radio-active substance.
3. Number of defects in the insulation of a hundred metre length of wire.

The Poisson distribution is defined by

$$P(X = r) = \frac{\lambda^r e^{-\lambda}}{r!}$$

where the range of possible values (r) is the set of non-negative integers $\{0, 1, 2, \dots\}$.

Poisson distribution has mean λ and its variance is also λ . It is the only distribution known so far, of which the mean and variance are equal. Poisson distribution possesses only one parameter (λ)

Example 27. A business firm receives on an average 2.5 telephone calls per day during the time 10.00 – 10.05 a.m. Find the probability that on a certain day, the firm receives (a) no call, and (b) exactly four calls, during the same period assume Poisson distribution and take $e^{-2.5} = 0.0821$.

Solution. Let the random variable X denotes the number of telephone calls received during the period 10.00 – 10.05 a.m. Here $\lambda = 2.5$. By Poisson distribution

$$P(X = r) = \frac{\lambda^r e^{-\lambda}}{r!}$$

(a) Probability of no call during the period

$$P(X = 0) = e^{-2.5} (2.5)^0 = e^{-2.5} = 0.0821$$

(b) Probability of exactly four calls during the period

$$P(X = 4) = e^{-2.5} (2.5)^4 / 4! = 0.1336$$

Example 28. A book contains 100 misprints distributed randomly throughout its 100 pages. What is the probability that a page observed at random contains at least 2 misprints. (Take $e = 2.718$)

Solution. Here $\lambda = np = 1/100 \times 100 = 1$

From the Poisson Distribution, we get

$$P(X = r) = \frac{\lambda^r e^{-\lambda}}{r!}$$

The probability that a page contains at least 2 misprints is $P(X \geq 2)$. But

$$\begin{aligned} P(x \geq 2) &= P(2) + P(3) + \dots = 1 - [P(0) + P(1)] \\ &= 1 - [1/e + 1/e] = 1 - 2/e = 0.2642 \end{aligned}$$

Poisson Distribution as an Approximation of the Binomial Distribution

The Poisson and Binomial distributions are related. The Poisson distribution can be a reasonable approximation of the Binomial, but only under certain conditions. Consider a Binomial distribution in which n trials take place and the probability of success is p . If n increases and p decreases such that np is constant, the resulting Binomial distribution can be approximated by a Poisson distribution with $\lambda = np$

To illustrate the above point. Table 15.2. Lists the probabilities for Binomial and Poisson distributions with $n = 15$, $p = 0.05$ and hence $\lambda = 15(0.05) = 0.75$. The remaining probabilities are all almost 0.

Table 15.2. The probabilities for binomial and Poisson distributions.

	Binomial $P(X = r); n = 15, p = 0.05$	Poisson $P(X = r); \lambda = 0.75$
$r = 0$	0.46339	0.47237
$r = 1$	0.36576	0.35427
$r = 2$	0.13475	0.13285
$r = 3$	0.03073	0.03321
$r = 4$	0.00485	0.00623
$r = 5$	0.00056	0.00090
$r = 6$	0.00005	0.00007
$r = 7$	0.00000	0.00001

Hypergeometric Distribution

Occasionally, the assumptions of the Poisson or Binomial can not be met even approximately. Subject only to the assumption of a random sample, the hypergeometric gives the probability of exactly r occurrence in n trials from a lot of N items having d defectives as

$$P(X = r) = \frac{C(d, r) C(N-d, n-r)}{C(N, n)}$$

The hypergeometric distribution is appropriate when independent trials are conducted, but the probability of occurrence of the event of interest changes from trial to trial.